

COURSE DETAILS

Faculty	: Information and Communication Technology
Programme	: Degree in Information Technology
	: Degree in Business Information Technology
	: Degree in Software Engineering with Multimedia
Year/ Semester	: Year 4 Semester 1
Lecturer	: Ms Ebiso & Ms Moopisa
Principal Lecturer	: Ms Mahlakeng
Pre-requisite	: None
Co-requisite	: None
Methods of Delivery	: Lectures
Credits	: 12
Contacts hours	: 3 (2 hours lecture + 1 Hour Tutorial)
No. of weeks	: 17 weeks long Semester (12 teaching, 1 midterm, 1 Non-Contact 1 Revision, 2 Final Exams)
No. of assignments	: 2
No. of exams	: 1
Portfolio	: None
Venue	:
Day	:
Time	:

Prepared by : Ms. Moopisa & Ms. Ebiso _____ Checked by: Ms Mahlakeng _____
 Signature :  Date: 05/08/2025 Signature:  Date 08/2025

This document comprises the following:

- Essential Information
- Specific Course Information
- Course Rules & Regulations
- Grades
- Plagiarism
- Course Introduction
- Course Aims & Objectives
- Learning Outcome
- Specific Generic Learning Skills
- Syllabus + Lecture Outline
- References
- Assignment Schedule
- Assessment Criteria
- Specific Criteria

Other documents as follows will be issued to you on an ongoing basis throughout the semester:

- Handouts for Assignments
- Submission Requirements + Guidelines

1.0 ESSENTIAL INFORMATION

- All modules other than electives are 'significant modules'
- As an indicator of workload one credit carries and additional 2 hours of self study per week. For example, a module worth 3 credits require that the student spends an additional 6 hours per week, either reading, completing the assignment or doing self-directed research for that module.
- **Submission of ALL assignment work is compulsory in this module, failure to do so, a DNS (Did not submit) grade would be awarded. An overall grade of DNC (Did not complete) would be awarded for those who fail to submit a major piece of assessment work (major assignment) or sit for either the midterm examination or final examination. A student cannot pass this module without having to submit ALL assignment work by the due date or an approved extension of that date.**
- All assignments are to be handed on time on the due date. Students will be penalised 10 percent for the first day and 5 percent per day thereafter for late submission (a weekend or a public holiday count as one day). Late submission, after the date Board of Studies meeting will not be accepted.
- Due dates, compulsory assignment requirements and submission requirements may only be altered with the consent of the majority of students enrolled in this module at the beginning/early in the program.
- Extensions of time for submission of assignment work may be granted if the application for extension is accompanied by a medical certificate.
- Overseas travel is not an acceptable reason for seeking a change in the examination schedule.
- Only the Head of School can grant approval for extension of submission beyond the assignment deadline.
- Re-submission of work can only receive a 50% maximum pass rate.
- Supplementary exams can only be granted if the level of work is satisfactory **AND** the semester work has been completed.
- Harvard referencing and plagiarism policy will apply on all written assignments.

2.0 SPECIFIC MODULE INFORMATION

- Attendance rate of 80% is mandatory for passing module.
- All grades are subject to attendance and participation.
- Absenteeism at any scheduled presentations will result in zero mark for that presentation.
- Visual presentation work in drawn and model form must be the original work of the student.
- The attached semester program is subject to change at short notice.

3.0 MODULE RULES AND REGULATIONS:

Assessment procedure:

- These rules and regulations are to be read in conjunction with the UNIT AIMS AND OBJECTIVES
- All assignments/projects must be completed and presented for marking by the due date.
- Marks will be deducted for late work and invalid reasons.
- All assignments must be delivered by the student in person to the lecturer concerned. No other lecturer is allowed to accept students' assignments.
- All tests/examinations are compulsory.
- Students must sit the test/examination on the notified date.
- Students are expected to familiarize themselves with the test/examination timetable.
- Students who miss a test/examination will not be allowed to pass.
- Any scheduling of tutorials, both during or after lecture hours, is TOTALLY the responsibility of each student. Appointments are to be proposed, arranged, confirmed, and kept, by each student. Failure to do so in a professional manner may result in penalty of grades. Tutorials WITHOUT appointments will also NOT be entertained.
- Note that every assignment is given an ample time frame for completion. This, together with advanced information pertaining deadlines gives you NO EXCUSE not to submit assignments on time.

4.0 GRADES

All modules and assessable projects will be graded according to the following system. With respect to those units that are designated 'Approved for Pass/Fail' the grade will be either PA or F:

<i>Grade</i>	<i>Numeric Grade</i>	<i>Description</i>
90 – 100	A+	Pass with Distinction
85 – 89	A	
80 – 84	A-	
75 – 79	B+	Pass with Credit
74 – 70	B	
65 – 69	B-	
60 – 64	C+	Pass
55 – 59	C	
50 – 54	C-, PX, PC	
0 – 49	F	Fail
	EXP	Exempted
	PC	Pass Conceded
	PP	Pass Provisional with extra work needed
	PX	Pass after extra work is given and passed
	X	Ineligible for assessment due to unsatisfactory attendance
	D	Deferred
	W	Withdraw
	DNA	Did Not Attend Module
	DNC	Did Not Complete Module

5.0 PLAGIARISM, COPYRIGHT, PATENTS, OWNERSHIP OF WORK: STUDENT MAJOR PROJECT, THESES & WORKS

5.0.1 Academic Dishonesty

Academic Dishonesty or academic misconduct is any type of cheating that occurs in relation to a formal academic exercise. It can include:

Plagiarism: The adoption or reproduction of original creations of another author (person, collective, organization, community or other type of author, including anonymous authors) without due acknowledgement.

Fabrication: The falsification of data, information or citations in any formal academic exercise.

Deception: Providing false information to an instructor concerning a formal academic exercise – e.g. giving a false excuse for missing a deadline or falsely claiming to have submitted work

Cheating: Any attempt to give obtain assistance in a formal academic exercise (like an examination) without due acknowledgement.

Bribery: or paid services. Giving certain test answers for money.

Sabotage: Acting to prevent others from completing their work. This includes cutting pages out of library books or wilfully disrupting the experiments of others.

Students involved in academic dishonesty will be treated as follows:

1. **Plagiarism and fabrication in Assignments, Projects or Presentations:** A zero (0) mark will be awarded to the work.
2. **Deception:** 10 % will be deducted on the first day of late submission and 5% a day thereafter. Late submission on or after BOS will not be accepted.
3. **Cheating:** Any form of assistance in a test will be rewarded a zero (0) mark. While in an examination the student automatically repeats the module.
4. **Bribery, Sabotage or Professional Misconduct:** Committing any of these offenses will lead to discontinuity from the University for a certain period depending on the seriousness of the offence.

5.0.2 Copyright, Patents and Ownership of Work

All use of the borrowed work, whether text or images must be referenced and acknowledged. Limkokwing is not liable for any infringement of copyright within students' work. Students are to abide by International Copyright laws. Students engaged in a project activity will be required, as a condition of acceptance as a student, to agree to assign to Limkokwing their right, title and interest in any invention or visual work arising from their studies with the University. Limkokwing has the right to keep students work for use in archives, exhibitions and events. Where students wish to keep original work, they must apply in writing and gain permission from the Academic Management.

5.0.3 Progression Policy

Students can only enroll into a higher semester if they have passed all courses in the preceding semester or have failed not more than two courses. A student cannot register for any course at a higher level if they fail its lower level unit which in essence is a prerequisite. If a student CGPA is less than 2.0, they could not proceed to the next semester.

6.0 MODULE INTRODUCTION

This is a course to provide a basic understanding of software project management concepts and the complex dimensions that make the management of software projects different from other types of projects.

7.0 MODULE AIMS AND OBJECTIVES

The module is intended to provide students with

- principles of software project management,
- Good knowledge of responsibilities of project manager.
- familiarity with different methods and techniques used for project management, project planning, project scheduling and project monitoring and be able to apply the knowledge in an effective manner.

8.0 LEARNING OUTCOME

Upon completion of the course, students will:

- Explain the concept and terminologies of software project management
- Explain principles of the project lifecycle
- Perform project planning, scheduling, monitoring & controlling
- Propose risk, resources, and configuration management plans
- Choose relevant and appropriate project management techniques for IT projects to initiate, plan, execute and evaluate a project
- Work in teams to create a project plan for a project scenario that includes key tasks, critical path, dependencies and a realistic timeline.
- Critically evaluate and discuss the issues around project management and its application in the real world with course participants and learners

9.0 SPECIFIC GENERIC LEARNING SKILLS

Upon completion of the module, student will acquire skills in:

- Preparing software cost estimates
- Preparing and monitoring project schedules
- Present strategies for gaining confidence in managing projects through simple project planning examples

10.0 UNIT SYLLABUS + LECTURE OUTLINE:

Week: 1

LECTURE 1:

Lecture Synopsis:

INTRODUCTION TO SOFTWARE PROJECT MANAGEMENT

- What is a Project?
- What is Management?
- Project Management
- Software Project Management Functions
- Software project (+++ Classifications) vs other project types
- Activities in Software Project Management
- Software Development Lifecycle Models
- Project success and failure
- Problems with Software Projects

Week:

2

LECTURE 2:

Lecture Synopsis:

SOFTWARE PROJECT EVALUATION AND PROJECT MANAGEMENT

- Project Evaluation
- Types of Assessments
 - Strategic
 - Technical
- Risk Evaluation
- Evaluation Techniques

Week:

3

LECTURE 3:

Lecture Synopsis:

SOFTWARE PROJECT PLANNING

- Project Planning, Scope and Objectives
- Project Feasibility Study
- Work Breakdown Structure
- Technical Planning
- Activity Planning
- Risk Reduction Planning
- Quality Planning
- Resource Planning

Week:

4

LECTURE 4:

Lecture Synopsis:

SOFTWARE EFFORT ESTIMATION

- Where are Estimates done
- The Basis for Software Estimating
- Estimating Effort, Duration & Resources (Time & Cost)
- Approaches to Estimation
- Problems and Uncertainty of Estimations
- Estimation Techniques

Class Test

Week:

5

LECTURE 5:

Lecture Synopsis:

SOFTWARE PROJECT SCHEDULING

- Objectives of Activity Planning
- When to Plan
- Project Schedules
- Sequencing and Scheduling Activities
- Network Planning Models
 - PERT
 - CPM
 - AoN (Activity on Node)
 - AoAN (Activity on Arrow Network)

Week:	6	
LECTURE 6:	RISK MANAGEMENT	
<i>Lecture Synopsis:</i>		○ Processes of Risk Management ○ Categories of Risks ○ Nature of Risk ○ Framework for Dealing with Risk ▪ Risk Identification ▪ Risk assessment/ analysis and prioritization ▪ Risk Planning ▪ Risk monitoring
Week:	7	MID-TERM EXAMINATION WEEK
Week:	8	NON - CONTACT WEEK
Week:	9	GRADUATION WEEK
Week:	10	
LECTURE 7:	RESOURCE ALLOCATION	
<i>Lecture Synopsis:</i>		○ Introduction ○ Nature of Resources ○ Identifying Resource Requirements ○ Resource Scheduling ○ Resource Allocation ○ Methods used in allocating and scheduling resources ○ Resource smoothing ○ Resource clashes ○ Publication of schedules
Week:	11	
LECTURE 8:	MANAGING SOFTWARE DEVELOPMENT PROJECTS	
<i>Lecture Synopsis:</i>		○ Project Life Cycles ○ From Traditional to Agile ○ Agile Methodologies ▪ Extreme Programming ▪ Dynamic System Development Method ▪ Scrum ▪ Kanban
Week:	12	
LECTURE 9:	MONITORING AND CONTROL	
<i>Lecture Synopsis:</i>		<i>Slides</i> ○ Project Control Cycle ○ Creating the Framework ○ Assessing/visualizing Progress ○ Collecting the Data ○ Cost Monitoring ○ Earned value analysis ○ Project Control

Week: 13
LECTURE 10: SOFTWARE QUALITY AND TESTING
Lecture Synopsis:

- Defining Software quality in project planning
- Importance of software quality
- Quality Assurance and Standards
 - Software quality Assurance Plan
 - Formal approaches (clean room)
- Quality Measures (FURPS...)
- Techniques to help enhance software quality
- Testing
 - Validation
 - Verification
- Quality plans

Week: 14
PRESENTATION: MAJOR ASSIGNMENT PRESENTATION

Week: 15
REVISION WEEK

Week: 16 & 17
FINAL EXAMINATION WEEK

11.0 REFERENCES

Recommended:

Software-Project-Management Bob Hughes Mike-Cotterell Rajib-Mall 6Ed

Additional:

- Villafiorita, A., Introduction to Software Project Management, CRC Press, 2014
- Robert. K. W., Effective Project Management Traditional, Agile, Extreme, Hybrid. John Wiley & Sons, Inc. Eighth Edition, 2019.
- Murray, P. A., the complete software project manager mastering technology from planning to launch and beyon. John Wiley & Sons, Inc, 2016
- Dimitrov, D., software project estimation intelligent forecasting, project control, and client relationship management., Apress, 2020

12.0 ASSESSMENT SCHEDULE

ASSIGNMENT DESCRIPTION	ISSUE DATE	DUE DATE	
WEEKLY PARTICIPATION (Lecture 1-10)	WEEKLY	EACH WEEK	10%
CLASS TEST 1: (Chapter 1-3)	WEEK 4	WEEK 4	10%
GROUP PROJECT	WEEK 3	WEEK 12	20%
MID-TERM EXAMINATION:	WEEK 7	WEEK 7	20%
FINAL EXAMINATION:	WEEK 16	WEEK 16	40%
TOTAL			100%

13.0 ASSESSMENT CRITERIA

Process of grading and criteria used to determine the grades, passes and high distinctions.

14.0 SPECIFIC CRITERIA

- Each assignment will be handed out with the project brief and will vary, depending on the teaching and learning objectives of the specific assignment.
- Each student will receive a completed assessment sheet back with their marks, thereby giving student feedback on each set criterion and the project as a whole.
- Marks for each project will be posted on the Bulletin Board with student number within 2 weeks of hand-in date.